RIGHT ORTHOGONALITY TO ONE OBJECT EXTENDS TO ITS THICK ENVELOPE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11260-11267, immediately following the statement at 11250-11258.

Editorial note. The induction is on the thickness degree n of the envelope; the base case uses all shifts at once.

Allowed standard prerequisites (editorial).

- Representable homological functors. Lemma lemma-representable-homological of this chapter, cited at proof lines 11264-11266. Hom(-, K) is homological, so vanishing propagates through distinguished triangles.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let E, K be objects of $\mathcal{D}$. The following are equivalent*

(1) $\mathrm{Hom}(E, K[i]) = 0$ *for all* $i \in \mathbf{Z}$,
(2) $\mathrm{Hom}(E', K) = 0$ *for all* $E' \in \langle E \rangle$.

Proof. The implication (2) $\Rightarrow$ (1) is immediate. Conversely, assume (1). Then $\mathrm{Hom}(X, K) = 0$ for all X in $\langle E \rangle_1$. Arguing by induction on n and using Lemma 0149 we see that $\mathrm{Hom}(X, K) = 0$ for all X in $\langle E \rangle_n$. □

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